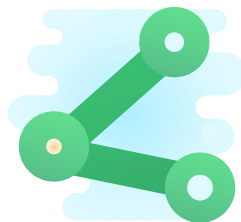


Patuakhali Science and Technology University

Faculty of Computer Science and Engineering



Network Routing and Switching CCE-416

Project Proposal

Multi-Site Hosting Provider Network

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Multi-Site Hosting Provider Network

1. Introduction

Modern web infrastructure requires consistent uptime, minimal latency, and zero data loss. Hosting providers running Virtual Private Servers (VPS), web servers, authoritative DNS, and Software-as-a-Service (SaaS) products face strict Service Level Agreements (SLAs). Single-point-of-failure architectures, flat network designs, and unsegmented traffic models expose cloud hosts to costly outages, lateral malware movements, and routing bottlenecks [1].

The infrastructure spans five distinct operational facilities: a primary Data Center (DC), a Head Office & Network Operations Center (HO/NOC), two Regional Points of Presence (PoP-1 and PoP-2), and a Disaster Recovery (DR) site.

By introducing hierarchical core-distribution-access switching, multi-area OSPF routing, dual-stack IPv4/IPv6 addressing, first-hop gateway redundancy (HSRP/VRRP), 802.1Q VLAN segmentation, EtherChannel bandwidth aggregation, strict Access Control Lists (ACLs), and Quality of Service (QoS) queues, the proposed network addresses existing scaling limitations while ensuring rapid failover and robust perimeter defense [2].

2. Project Objectives & Scope

2.1. Objectives

1. To build a reliable, high-speed network infrastructure for hosting virtual servers, web applications, and domain services across multiple locations.
2. To design a clear multi-site network structure that connects the head office, data center, regional branch offices, and backup recovery site.
3. To implement both IPv4 and IPv6 network addressing schemes across all connected devices.
4. To separate internal network traffic into distinct virtual groups to keep data organized and prevent unauthorized access.
5. To isolate public-facing websites and DNS servers in a secure public area while keeping internal databases protected.
6. To enforce network security rules, remote login controls, and port restrictions to block external threats.
7. To prioritize time-sensitive traffic like voice calls and core business software over bulk background backups.
8. To test and verify network stability by simulating cable cuts, hardware failures, and unexpected outages.
9. To produce complete network topology diagrams, setup scripts, and empirical test reports for verification.

2.2. Project Scope & Boundary

The scope covers network layers 1 through 4, along with essential network services (DNS, DHCP relay, Web hosting, SSH). Cisco packet tracer will be initially used for emulation, while GNS3 will be used for advanced routing and security testing. The project does not include physical cabling, server OS configuration, or application-level software deployment.

3. Proposed Network Architecture

The architecture mirrors a production hosting provider infrastructure, organized into five physical and logical sites connected through redundant WAN and fiber backbones.

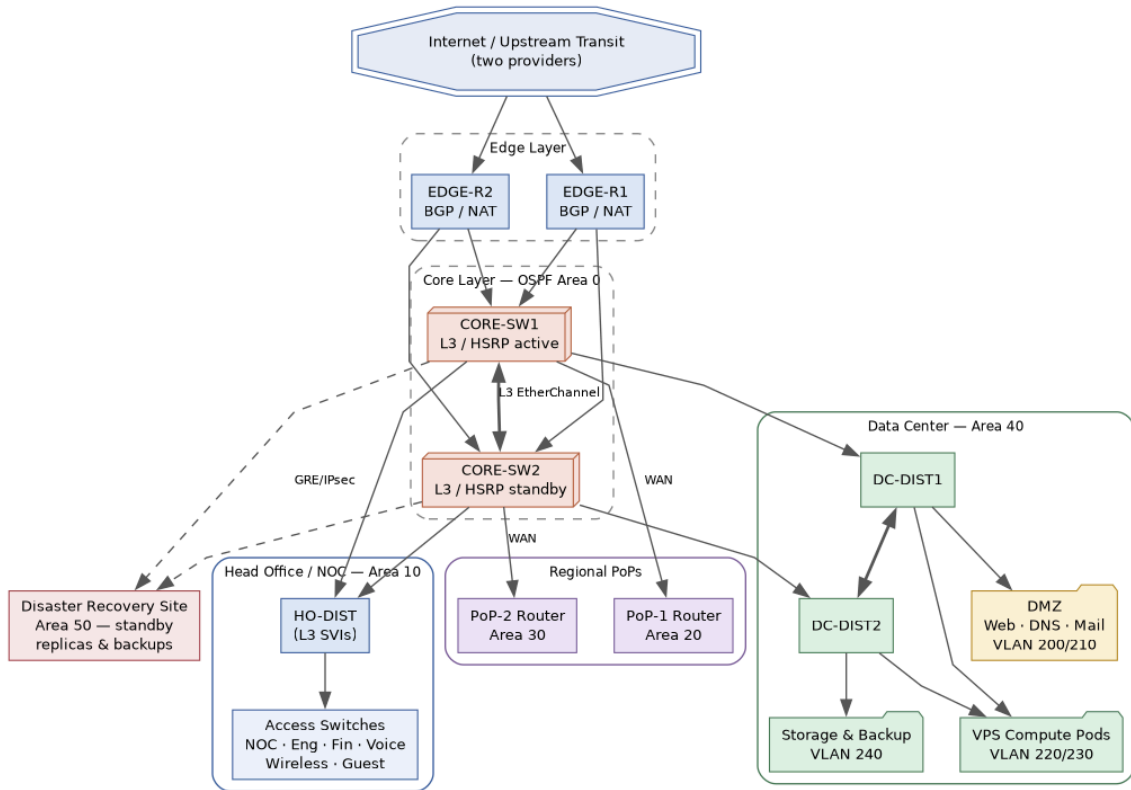


Figure 1: High-Availability Multi-Site Hosting Provider Network Architecture

3.1. Site Breakdown & Physical Distribution

Site Name	Minimum Devices	Primary Role & Hosted Services
Head Office (HO)	2 Core/Dist switches, 4 Access switches	Corporate NOC, IT, HR, Finance, Voice, Wireless, Guest Access
Data Center (DC)	2 Core Routers, 2 Dist Switches, 4 Rack Switches	VPS Compute Pods, Web Server Pools, Any-cast DNS, DMZ, Storage Array
PoP-1 (Regional)	1 Edge Router, 2 Access Switches	Regional Edge Caching, Customer VPS Edge Access, Local Transit
PoP-2 (Regional)	1 Edge Router, 2 Access Switches	Secondary Regional Edge Caching, Customer Aggregation
Disaster Recovery (DR)	1 DR Router, 2 Storage Switches	Offsite Backup Replicas, Standby Database & DNS Mirrors

3.2. Hierarchical Layer Design

1. Core Layer: High-speed Layer-3 switching backbone running OSPF Area 0. Transports traffic between DC, HO, and PoP aggregators without processing CPU-intensive ACL filtering.

2. Distribution Layer: Implements inter-VLAN routing (SVIs), HSRP/VRRP gateway redundancy, OSPF route summarization, and security ACL execution.
3. Access Layer: Connects physical servers, hypervisors, NOC workstations, and IP phones. Enforces Layer-2 security (Port Security, BPDU Guard, Storm Control).
4. Edge / Perimeter: Dual-homed BGP routers (EDGE-R1 and EDGE-R2) interfacing with Tier-1 upstream transit providers, executing Stateful NAT/PAT and Edge ACL filtering.

4. IP Addressing

The hoster is assigned the private address space 10.0.0.0/8. Using Variable Length Subnet Masking (VLSM), subnets are allocated hierarchically by location to facilitate summary routes at ABR boundaries.

Location	IPv4 Subnet Block	Mask	Subnet Purpose
Head Office (HO)	10.10.0.0/16	255.255.0.0	NOC, Administration, Corporate Departments
PoP-1 (Regional)	10.20.0.0/16	255.255.0.0	Regional Customer Edge & Transit
PoP-2 (Regional)	10.30.0.0/16	255.255.0.0	Secondary Customer Edge
Data Center (DC)	10.40.0.0/16	255.255.0.0	DMZ, VPS Pods, Web Hosting, Storage
Disaster Recovery (DR)	10.50.0.0/16	255.255.0.0	Backup Servers & Mirror Replicas
WAN Interconnects	10.254.0.0/24	255.255.255.0	Point-to-point router links (/30 subnets)

5. Network Security

The security architecture follows a Defense-in-Depth strategy, using perimeter firewalling, segment isolation, and access control matrices.

5.1. Access Rules & Enforcement Matrix

Rule	Security Directive	Enforcement Mechanism	Applied Interface
1	Guest users isolated from internal servers	Deny traffic from 10.10.90.0/24 to internal networks	Inbound on VLAN 90 SVI
2	General users restricted from Finance VLAN	Deny 10.10.60.0/23 to 10.10.30.0/24 except HTTP billing API	Inbound on VLAN 60 SVI
3	Remote management restricted to IT NOC	Permit SSH (TCP 22) only from 10.10.40.0/24 to MGMT VLAN 100	VTY Lines on all devices

4	Public hosting services isolated in DMZ	Stateful inspection filtering external requests to DMZ Web/DNS	DC Edge Router / DMZ SVI
5	Voice & critical SaaS traffic permitted	Explicit permit rules for VoIP SIP (5060) and RTP traffic	Core QoS / ACL policies
6	Full documentation & rule testing	Logging enabled on implicit deny statements	All ACL configurations

6. Essential Network Services

1. Authoritative & Recursive DNS: Deployed across two redundant nodes (10.40.210.10 primary in DC, 10.50.210.10 secondary in DR) offering public name resolution for hosted tenant domains.
2. Web Server Pools: Scalable Apache/Nginx web server clusters hosted in DMZ VLAN 200, load-balanced via virtual VIPs.
3. Centralized DHCP & DHCP Relay: A central redundant DHCP server (10.40.70.15) assigns dynamic IPs for NOC users, Voice devices, and temporary VPS staging pods. Switches running SVIs use ip helper-address 10.40.70.15 to forward requests.
4. NAT / PAT (Port Address Translation): Edge routers execute overload NAT to map internal corporate subnets (10.10.0.0/16) to public IP addresses while protecting internal topology structure.

7. Project Timeline (Gantt Chart)

Project Task	W1-2	W3-4	W5-6	W7-8	W9	W10	W11	W12
Requirements & Topology Design	X							
IPv4/VLSM & IPv6 Address Design		X						
Switching, VLANs & EtherChannel			X					
Multi-Area OSPF & Gateway Redundancy				X				
Security, Extended ACLs & SSH					X			
Hosting Services (DNS, DMZ, NAT)						X		
Failure Injection & Convergence Testing							X	
Final Technical Report & Defense								X

8. Conclusion

This proposal presents a comprehensive, high-availability network design tailored specifically for a multi-site cloud hosting provider. By incorporating 3-tier hierarchical architecture, dual-stack IPv4/IPv6 addressing, multi-area OSPF routing, HSRP/VRRP gateway redundancy, and defense-in-depth security filtering, the proposed design guarantees operational resilience, scalable tenant isolation, and strict SLA compliance.

9. References

- [1] Uptime Institute, “Annual Outage Analysis.” [Online]. Available: <https://uptimeinstitute.com/resources/research-and-reports>
- [2] D. Teare, B. Vachon, and R. Graziani, *Implementing Cisco IP Routing (ROUTE): Foundation Learning Guide*. Cisco Press, 2015.